Toward a Low-Latency Multimodal Full-Duplex Model

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1 Motivation

Full-duplex spoken language models (FD-SLM) have become a popular research topic in recent years, as their concurrent input and output audio channels allow us to model more complex conversational scenarios. For example, an FD-SLM can handle overlapping speech, respond to interruptions at any time, or produce backchannels, such as by using pauses like "hmm" or "um" (Défossez et al., 2024). Recently, full-duplex models have been able to understand visual information in addition to speech, which can provide additional context to the model(Cui et al., 2026). Vision is particularly important to conversational scenarios, as natural human interaction or conversation contains an array of visual elements such as gestures and facial expressions. Although multimodal full-duplex models (MFDMs) such as MiniCPM-o4.5 (Cui et al., 2026) and Realtime-Venus (Zhao et al., 2026) achieve streaming full-duplex multimodal interaction, these leading models have end-to-end latency exceeding 1s, which is too long for natural real-time interaction. So, our goal is to build upon D2(Chang et al., 2026), a successful low-latency FD-SLM, by introducing visual information so that it understands visual cues while retaining its improved latency.

2 Related Work

Low Latency Full-Duplex Spoken Language Models Moshi (Défossez et al., 2024) introduced a real-time full-duplex spoken language model capable of simultaneously processing user speech and generating responses. Unlike traditional speech systems that separate recognition, language generation, and speech synthesis, Moshi models incoming and outgoing audio as parallel streams. This allows it to handle interruptions and overlapping speech without requiring explicit turn boundaries. However, Moshi does not incorporate visual information, limiting its ability to respond to changes in the user's environment. Our work builds on the low-latency full-duplex paradigm demonstrated by Moshi while extending it to audiovisual interaction.

D2(Chang et al., 2026) builds on the approach in Moshi and other FD-SLMs, but innovates by starting with an already capable half-duplex model in Qwen3-Omni(Xu et al., 2025). So, D2 does not need to train the full multimodal model from scratch, instead using distillation to train a causal and streaming version of the encoder in Qwen3-Omni. Additionally, the model operates in 80-ms intervals similarly to Moshi, allowing it to achieve more efficiency and a reasonable latency. Fine-tuning is done on synthetic full-duplex scenarios, allowing the model to learn how to listen, stop, interrupt, etc. The result is a model that improves on Moshi in both interruption success and latency, as well as on benchmarks such as VoiceBench(Chen et al., 2026), establishing a new state of the art for latent full-duplex models.

High Latency Multimodal Full-Duplex Models Recent multimodal full-duplex models have extended real-time interaction to simultaneous audio and visual understanding. MiniCPM-o 4.5 (Cui et al., 2026) supports streaming audio and video inputs while generating speech responses, enabling interaction with continuously changing visual scenes. Similarly, Realtime-Venus (Zhao et al., 2026) integrates audiovisual perception with full-duplex dialogue and asynchronous task execution. These systems demonstrate the feasibility of real-time multimodal interaction, but incorporating continuous visual information creates additional computational demands. Our approach differs by extending the existing low-latency D2 architecture rather than training a new multimodal full-duplex model from scratch. By reusing the pretrained vision capabilities of Qwen3-Omni, we aim to achieve audiovisual understanding while minimizing additional inference latency.

3 Method

D2 successfully transforms Qwen3-Omni into a full-duplex spoken language model, but it fails to utilize the vision ability of Qwen3-Omni. We will introduce the vision encoder from the original Qwen3-Omni into the D2 architecture, and insert vision tokens into the streaming input chunks to achieve full-duplex speech and vision understanding. To do this, we proceed similarly to D2 by using a causal encoder built via representation distillation from Qwen3-Omni, and use the same 80-ms intervals. Then, we will construct fully synthetic full-duplex audio-video conversations from existing question and answer datasets to fine-tune the model.

Our implementation will follow two stages. First, we will adapt the pretrained vision encoder to produce compact representations suitable for streaming inference, using representation distillation if necessary. We will initially test whether the existing vision encoder can process frames incrementally with acceptable latency before introducing additional training. Second, we will integrate these representations into D2 and perform supervised fine-tuning on synchronized audiovisual conversations. This approach allows us to reuse pretrained multimodal capabilities while adapting the model to continuous interaction.

Dataset We will use the AVQA training split to construct synthetic audiovisual question-answering conversations for multimodal supervised fine-tuning. Since AVQA consists of multiple-choice questions, we will use other models that can understand videos to paraphrase the choices into more diverse scenarios. Also, to maintain normal user-assistant chat ability, we will use Instruct2s-200k for the audio source, along with a placeholder video, such as a person facing the webcam to fill the vision token slot. For both datasets, we will use the Kokoro TTS model to synthesize the assistant voice, so that we can have a consistent voice template across training.

End-to-End SFT To make audio and vision have an aligned timeline, we will downsample the input so that the vision encoder generates 12 tokens per second and uses one placeholder every 24 tokens to achieve a 12.5 Hz update rate. During training, we will encode audio and video, aligning them in chunks of `[user audio, self audio, vision token, self text]`, and do supervised fine-tuning just like in D2.

Inference See Algorithm 1

Algorithm 1 D2-Omni Inference

Require: Streaming audio encoder E, vision encoder E_v, dialogue model M, frame duration Δ

Initialize model context H
$S_0 \leftarrow \text{Silence}(\Delta)$
$B \leftarrow \emptyset$, $t_{\text{next}} \leftarrow 0$
for each frame $i = 1, 2, \ldots$ **do**
  if $(i-1)\Delta \geq t_{\text{next}}$ **then**
    $B \leftarrow E_v(\text{current video frame})$
    $t_{\text{next}} \leftarrow t_{\text{next}} + 1\,\text{s}$
  end if
  Observe user audio X_i of duration Δ
  $U_i \leftarrow E(X_i)$ ▷ One user embedding
  $A_i \leftarrow E(S_{i-1})$ ▷ One self embedding
  $V_i \leftarrow \text{PopNext}(B)$ ▷ Next visual token, or empty
  $H \leftarrow \text{Prefill}_M(H, [U_i; A_i; V_i])$
  $(z_i, y_i, H) \leftarrow \text{GenerateFrame}_M(H)$ ▷ One text token and one speech-code frame
  Emit text token z_i
  $S_i \leftarrow \text{Render}(y_i)$
  Emit S_i
end for

4 Evaluation

We will evaluate our model along three dimensions: audiovisual understanding, full-duplex conversational performance, and inference latency. Our primary goal is to demonstrate that adding streaming vision improves multimodal understanding without substantially degrading D2's existing speech capabilities or responsiveness.

Audiovisual Understanding We will evaluate our model on the held-out AVQA benchmark (Yang et al., 2022), which contains questions requiring information from both audio and video. We will report answer accuracy and compare against the original audio-only D2 model to measure the benefits of incorporating visual information. We will also evaluate our model with visual inputs removed to determine whether it effectively uses the additional modality. If time permits, we will evaluate video understanding separately using Video-MME (Fu et al., 2025).

Full-Duplex Conversational Performance We will use VoiceBench (Chen et al., 2026) to evaluate whether the model retains its original speech understanding capabilities. We will also measure turn-taking success, interruption success, and response latency using the evaluation methodology from D2. Comparing these results against the original D2 model will allow us to identify degradation caused by multimodal integration.

Inference Efficiency We will measure end-to-end response latency, inference time, and real-time factor (RTF) using a fixed hardware configuration. We will compare our model against D2 and, where feasible, MiniCPM-o 4.5 under comparable conditions. We will also investigate how different visual sampling rates affect accuracy and inference efficiency. Our goal is to achieve meaningful audiovisual understanding while maintaining real-time performance.

Finally, we will demonstrate the model in a live webcam conversation, where it must process incoming audio and video while generating spoken responses. Although understanding facial expressions and other subtle conversational cues would be valuable, we leave this more complex evaluation of multimodal conversational behavior to future work.

5 Feasibility

We generally believe that Qwen3-Omni-30B-A3B has excellent multimodal capability, and with the success of distilling the audio encoder in D2, we believe that a similar distillation of the model's visual encoder will let us incorporate visual information, especially with the help of supervised fine-tuning. In particular, other multimodal models like MiniCPM-o4.5 succeed in this domain, so we believe it is feasible to achieve similar or better performance, especially by improving latency, as this is a benefit of D2-based models.

From a computational perspective, it took 10 hours to train D2 with 4 RTX Pro 6000 on MIT's ORCD, which should be very feasible even if we include the vision tokens. For inference, it uses an MoE architecture and can perform real-time inference with an RTF of 0.6 on RTX Pro 6000. In our proposed D2-Omni, we will include a vision token in each input chunk, which will introduce 1.33x computation overhead, so achieving real-time inference may require an H100 or H200.

We will implement our model using PyTorch and the existing D2 codebase, with training performed on MIT's ORCD GPU cluster. Our work will proceed in four stages. First, we will integrate the pretrained vision encoder and test its streaming performance, performing representation distillation if necessary. Second, we will construct synthetic audiovisual training data using AVQA and speech instruction datasets. Third, we will perform supervised fine-tuning and verify that the model can process simultaneous audio-video inputs and generate speech. Finally, we will benchmark multimodal understanding, conversational performance, and inference latency.

The main challenges are aligning visual and audio representations and limiting the additional computation introduced by vision processing. If full multimodal fine-tuning proves too expensive, we will prioritize a working streaming audiovisual model using a frozen vision encoder and parameter-efficient fine-tuning. More advanced objectives, such as improving turn-taking decisions using facial expressions, will be considered only if time permits.

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